XIAOQI JIANG

pronounced *shyow-chee jyahng*

Data Science Institute, The University of Chicago
5460 S University Ave, Chicago, IL 60615, USA
xqj@uchicago.edu | +1 (872) 381-1760 | shawj1029.github.io

EDUCATION

The University of Chicago Expected Dec. 2026
M.S. in Data Science, Data Science Institute — GPA: 4.0/4.0 Chicago, IL, USA

Zhejiang University June 2025
B.S. in Physics, School of Physics — GPA: 3.88/4.0 Hangzhou, Zhejiang, China
Thesis advisor: Prof. Renyue Cen.

RESEARCH INTERESTS

I am interested in theoretical and computational astrophysics, especially galaxy formation and the baryon cycle across scales. I am excited about probing physical processes by building and analyzing numerical simulations.

RESEARCH EXPERIENCE

Turbulence Structures in the Multiphase ISM Aug. 2026 — present
Visiting Student, Princeton University; advised by Dr. Chang-Goo Kim and Dr. Rajsekhar Mohapatra

- Implement Fourier and finite-difference Helmholtz solvers for the shearing, stratified TIGRESS box.
- Show that a global decomposition leaves spurious, nearly cancelling modes in the warm and cold gas, and that masking the hot gas first reduces this.
- Measure the scale dependence of the compressive fraction in the warm and cold gas, and relate it to supernova feedback and star formation.

Thin Disks as a Function of Baryonic Mass Dec. 2025 — present
Graduate Researcher, advised by Prof. Andrey Kravtsov

- Built an image-processing pipeline for ALFALFA H I-selected galaxies in DESI Legacy Surveys r-band images, including removal of foreground and background contaminants, segmentation, and fit masks.
- Measured axis ratios with light moments and PSF-convolved Sérsic fits, and verified their agreement.
- Modeled the dependence of disk thickness on stellar mass, baryonic mass, and rotation velocity.

Planar Structures in ΛCDM Universes Nov. 2023 — Sept. 2026
Undergraduate/Graduate Researcher, advised by Prof. Renyue Cen, in collaboration with Prof. P. J. E. Peebles

- Placed Local Group analog observers in cosmological N-body simulations, and scanned stacks of thin slabs over millions of directions per observer in redshift space.
- Vectorized the halo counting as matrix projections onto all directions, and parallelized it across hundreds of CPU cores by splitting directions into blocks and distributing observers.
- Quantified how rare the local planes are in ΛCDM and how the observers who see them cluster.

The Quasar Dipole in Cosmological Simulations Winter 2026
Independent course project, Statistical Methods in Astrophysics

- Built full-sky mock quasar catalogs for Local Group-like observers from periodic lightcones, including redshift-space distortions and the kinematic dipole from the observer's motion.
- Fit dipoles on HEALPix maps and tested how often the observed amplitude and CMB misalignment occur in ΛCDM.

PUBLICATIONS

– **Jiang, X.**, Cen, R., & Peebles, P. J. E. "Planar structures in ΛCDM universes." *in prep.*
– **Jiang, X.** & Kravtsov, A. "Through thick and thin: disk shapes of dwarf galaxies as a function of baryonic mass." *in prep.*

TEACHING

Graduate Student Teaching Assistant — BIOS 21403: Quantitative Bioimaging Spring 2026
The University of Chicago, Biological Sciences Collegiate Division

- Led discussion sections, held office hours, and graded for a course on image analysis in Python.

CONFERENCES & PRESENTATIONS

Open SkAI 2025 — attendee Sept. 2025
27th Guo Shoujing Symposium on Galactic Cosmology — poster presenter June 2025

HONORS & AWARDS

Outstanding Student, Zhejiang University 2025
Excellent Graduate, Zhejiang University 2025
First Class Scholarship, Zhejiang University 2022

COURSEWORK

* graduate course

Mathematics
Approximation Algorithms*; Foundations of Machine Learning and AI*; Mathematical Analysis; Advanced Algebra and Analytic Geometry; Ordinary Differential Equations; Methods of Mathematical Physics; Probability and Statistics

Physics
Cosmology I*; Galaxies*; Statistical Methods in Astrophysics*; Quantum Mechanics; Electrodynamics; Theoretical Mechanics; Statistical Mechanics; Computational Physics; Galactic Astronomy

SKILLS

Programming: Python, R, C++, SQL, LaTeX, Bash
Computational: HPC (multi-core CPU and GPU), Slurm
Methods: Machine Learning, Monte Carlo, Bayesian Inference, MCMC, Fourier Analysis
Languages: Mandarin (native), English (fluent)